

Project Proposal: KAI 9

Autonomous Care Companion

1. Executive Summary & Mandate

Objective: To engineer KAI 9, a hybrid autonomous sanitation and social companion robot designed to address the dual crisis of aging demographics and social isolation. KAI 9 will serve as an independent living support system, combining robust mechatronic utility (vacuuming) with empathetic digital mediation (voice-first communication).

Scope: This 8-week project will deliver a fully functional, 3D-printed prototype meeting all core mandates: Autonomous Movement, Active Living Prompts, Digital Gateway Integration, and IBM Granite AI-assisted behavioral reasoning. Secondary features have been explicitly relegated to a Phase 2 rollout to guarantee the stability and grade-readiness of the core Minimum Viable Product (MVP).

2. System Architecture

Hardware Subsystem

A custom 3D-printed K9 chassis housing a high-torque track-and-motor drivetrain, an integrated vacuum assembly, and a robust power management system with a self-charging station. The sensory array includes LiDAR for spatial mapping, alongside a microphone, speaker, and camera for environmental and user interaction.

Software Subsystem

Built around a state-machine architecture handling autonomous navigation (obstacle avoidance via LiDAR), API integrations (WhatsApp, Spotify, Audible) for the digital gateway, and a Language Processor for voice commands.

Intelligence Layer (IBM Granite)

IBM Granite will act as an active co-pilot for developing software architecture, reasoning through sensor noise logic, and refining the behavioral dialogue to ensure user prompts are supportive rather than infantilizing.

Integration Point

A Raspberry Pi 5 will serve as the central compute unit, bridging the high-level Python AI/API logic with the low-level embedded hardware actuation.

3. Team Roles & Task Division

Tasks are strictly divided into Hardware (3 engineers) and Software (3 engineers) domains. The team will dynamically allocate these tasks based on immediate project bottlenecks.

Hardware Team Tasks

- Design and 3D print the K9 chassis (body, internal frame).
- Source, mount, and wire the track and motor drivetrain.
- Integrate the vacuum suction unit into the chassis.
- Mount the sensor array (LiDAR, Camera).
- Install the audio interface (Mic array and Speakers).
- Design and build the battery/power management system and self-charging station.
- Ensure physical robustness and safety fail-safes.

Software Team Tasks

- Develop autonomous movement and pathfinding logic (using LiDAR data).
- Program the movement decision-maker and obstacle avoidance.
- Integrate third-party APIs (WhatsApp, Spotify, Audible).
- Develop the Language Processor for voice interactions.
- Program the Wellbeing tracker and Physical Activity prompts.
- Implement the Emergency Protocol.
- Manage the IBM Granite AI co-pilot integration for code reasoning and behavioral design.
- Draft the System Architecture diagrams and Granite Usage Report.

4. 8-Week Project Timeline

We ek	Phase	Hardware Tasks	Software Tasks
1	Design & Sourcing	Finalize CAD; Order components; Prototype vacuum mechanism.	Setup dev environment; Establish AI co-pilot workflow; Begin API research.
2	Fabrication & Logic	Begin 3D printing chassis; Assemble internal frame.	Write core movement decision-maker logic; Setup sensor data pipelines.
3	Actuation & Processing	Mount motors and tracks; Test power management.	Integrate LiDAR data processing; Build basic obstacle avoidance.
4	Sub-System Assembly	Integrate vacuum and charging contacts; Mount Pi.	Develop Voice/Language processor; Integrate WhatsApp/Spotify APIs.
5	System Convergence	Wire sensor array (mic, speaker, camera) to compute unit.	Merge navigation logic with physical motors; Refine dialog prompts.
6	Behavioral Integration	Hardware debugging (fix torque/weight issues).	Implement Wellbeing tracking and Emergency Protocols.
7	Testing & Safety	Physical robustness testing; Failsafe testing.	Edge-case debugging; Finalize autonomous loop.
8	Deliverables	Final physical polish.	Record Demo Video; Compile Architecture/AI Usage/Ethics reports.

5. Budget & Resource Justification

Component Category	Expected Cost (£)	Justification / Notes
Compute & Vision	£95.00	Raspberry Pi 5 & Camera. (Provided by Department)
Navigation	£90.00	RPLiDAR A1 (or equivalent). Essential for complex pathfinding.
Drivetrain	£70.00	High-torque DC motors, encoders, motor drivers, tracks.
Power System	£50.00	LiPo battery, BMS board, buck converters.
Sanitation (Vacuum)	£45.00	Brushless impeller motor, filters, custom housing.
Chassis & Fabrication	£40.00	PETG/ABS filament for a robust structure.
Sensors & Audio	£30.00	Mic array, speakers.
Contingency	£40.00	10% buffer for blown components or failed prints.
Total System Value	£460.00	Represents the true cost of a functional, robust prototype.
Required Out-of-Pocket	£365.00	Even with the Raspberry Pi provided, the required remaining hardware strictly exceeds the £300 budget. A funding extension of £65+ is formally requested.

6. Evaluation & Risk Management

Alignment with Assessment Criteria

The project's success will be directly measured against the module's core metrics:

- **Hardware-Software Integration:** Proven by seamless vacuuming during autonomous navigation.
- **Meaningful Granite Usage:** Documented extensively in the final 2-3 page Granite Usage Report.
- **Safety & Ethics:** Local processing of user data and physical motor fail-safes.
- **Robustness:** Use of high-torque motors and thick-walled PETG printing to prevent fragile hardware.
- **Usability for Elderly:** Voice-first API integration; prioritizing cognitive ease.

Primary Risks

Risk	Details	Mitigation
Supply Chain Lead Times	Critical components (LiDAR, Tracks) require 10-15 day shipping.	Order placed in Week 1; software team will use simulation and dummy data for the first 14 days.
Budget Deficit	The £300 cap forces the use of cheaper, underpowered components.	Requesting immediate budget extension to secure robust drivetrain parts that will not stall under chassis weight.

7. Future Scope (Phase 2)

To ensure maximum stability for the core deliverables, the following features have been mapped as stretch goals or Phase 2 implementations:

- Advanced conversational abilities and voice calls.
- Logging events and learning user life routines.

- Alarm clock and medication reminders.
- Google API for calendar collaboration.
- Customisation (e.g., Son's voice as the set voice, barking at the mailman).
- Companion mobile application.